

Electric Potential

Energy per unit charge: the scalar that makes electric problems easier, and the 'voltage' behind every battery, nerve and X-ray tube.

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The electric force is conservative, like gravity, so it has a potential energy. Dividing that energy by the charge gives the **electric potential** V , the familiar “voltage”. Potential is a **scalar**, so it adds without any vector components, and the field can be recovered from it by differentiation. It is also what we actually measure and control: a battery maintains a potential difference, a nerve cell keeps a potential across its membrane, an X-ray tube accelerates electrons through one, and an ECG records potential differences across the body. This chapter builds the idea of potential and connects it with the field.

YOU WILL BE ABLE TO

- Calculate the electric potential energy of systems of point charges.
- Define potential and potential difference, and use volts and electron-volts.
- Calculate potentials of point charges and continuous distributions by superposition.
- Find potential from field ($\Delta V = - \int \vec{E} \cdot d\vec{l}$) and field from potential ($\vec{E} = -\nabla V$).
- Sketch equipotential surfaces and relate them to field lines.
- Apply potential to conductors, charged particles, nerve membranes and X-ray tubes.

THINK FIRST

A bird perches on a single high-voltage power line at 100 000 V and comes to no harm. A person touching the same line while standing on the ground would be killed. What is the difference?

What drives current through a body is a potential **difference** across it, not the potential itself. The bird's two feet are on the same wire, at almost exactly the same potential (the tiny resistance of a few centimetres of thick cable gives a difference of only millivolts), so essentially no current flows through the bird. A person touching the line while standing on the ground has 100 000 V between hand and feet. The bird is in danger only if it touches two wires at once, or a wire and a grounded pole, which is why large birds are sometimes electrocuted on poorly designed pylons.

§15.1

Electric potential energy

Two point charges q_1 and q_2 a distance r apart have potential energy

$$U = k \frac{q_1 q_2}{r},$$

taking $U = 0$ when they are infinitely far apart. This follows from integrating the work done against Coulomb's force, exactly as for gravity in Chapter 8. Here the sign follows the charges:

- **Like charges** have $U > 0$. Work must be done to push them together, and they release energy as they fly apart.
- **Unlike charges** have $U < 0$. They form a bound system, and energy must be supplied to separate them.

For a system of several charges, add the energy of **every pair**, counting each pair once:

$$U = k \sum_{i < j} \frac{q_i q_j}{r_{ij}}.$$

This is the work needed to assemble the system from charges that start infinitely far apart.

WORKED EXAMPLE 15.1

Assembling three charges

How much work is needed to bring three charges of $+2.0 \mu\text{C}$ each from far apart to the corners of an equilateral triangle of side 10 cm?

Solution

There are three pairs, each at a distance of 0.10 m:

$$U = 3 \frac{kq^2}{a} = 3 \frac{(8.99 \times 10^9)(2.0 \times 10^{-6})^2}{0.10} = 1.08 \text{ J.}$$

If the charges were released, this 1.08 J would become their kinetic energy as they flew apart.

§15.2

Electric potential

The **electric potential** at a point is the potential energy per unit charge that a test charge would have there:

$$V = \frac{U}{q_0}.$$

The unit is the **volt**: $1 \text{ V} = 1 \text{ J/C}$. The **potential difference** between points **a** and **b** is the work per unit charge that the field does when a charge moves from **a** to **b**, with a minus sign:

$$\Delta V = V_b - V_a = -\frac{W_{a \rightarrow b}}{q}.$$

A charge **q** moving through a potential difference ΔV changes its potential energy by

$$\Delta U = q \Delta V.$$

If only electric forces act, it gains kinetic energy $\Delta K = -q \Delta V$.

- **Positive** charges are pushed towards **lower** potential, like masses rolling downhill.
- **Negative** charges, such as electrons, are pushed towards **higher** potential.

The electron-volt. The energy gained by an electron (or any particle of charge **e**) accelerated through a potential difference of 1 V is

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J.}$$

It is the natural unit of energy for atoms (bond energies are a few eV), for X-rays (keV) and for nuclei (MeV).

WORKED EXAMPLE 15.2

Accelerating particles

(a) A proton starts from rest and is accelerated through 1.0 kV. Find its kinetic energy (in eV and J) and its speed. (b) An alpha particle (charge $+2e$) is accelerated through the same 1.0 kV. What kinetic energy does it gain?

Solution

(a) $K = e(1000 \text{ V}) = 1.0 \text{ keV} = 1.6 \times 10^{-16} \text{ J}$.

$$v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2(1.6 \times 10^{-16})}{1.67 \times 10^{-27}}} = 4.4 \times 10^5 \text{ m/s}.$$

(b) Twice the charge gains twice the energy: $K = 2e(1000 \text{ V}) = 2.0 \text{ keV}$.

Evaluate. Electron-volts make such bookkeeping trivial: the energy in eV is just the charge (in units of e) times the voltage.

§15.3

Potential of point charges and charge distributions

The potential of a point charge q at distance r is

$$V = \frac{kq}{r},$$

with $V = 0$ at infinity. It is positive near positive charges and negative near negative ones.

Potentials from several sources add as scalars:

$$V = k \sum_i \frac{q_i}{r_i}, \quad V = k \int \frac{dq}{r}.$$

This is usually far easier than adding field vectors.

Ring of charge (charge Q , radius R), on its axis at distance x . Every element is the same distance $\sqrt{x^2 + R^2}$ away, so

$$V = \frac{kQ}{\sqrt{x^2 + R^2}}.$$

No components are needed, because potential is a scalar.

Spherical shell or conducting sphere (charge Q , radius R):

- Outside: $V = kQ/r$.
- Inside: the field is zero, so V is constant and equal to its surface value, $V = kQ/R$.

WORKED EXAMPLE 15.3

Potential at the centre of a square

Charges of $+q$, $+q$, $-q$ and $-q$ sit at the four corners of a square of side a , with the two like charges on adjacent corners. Find the potential and the electric field at the centre.

Solution

Potential. All four charges are the same distance $a/\sqrt{2}$ from the centre, so their potentials cancel:

$$V = \frac{k(q + q - q - q)}{a/\sqrt{2}} = 0.$$

Field. The field is *not* zero. Each positive charge's field points away from it, and each negative charge's field points towards it. These add up to a net field pointing from the positive side of the square towards the negative side.

Lesson. $V = 0$ at a point does **not** mean $\vec{E} = 0$ there, and vice versa. The field depends on how V changes nearby, not on its value at that point.

§15.4

Potential from the field

The work done by the field on a charge q moving through a small displacement $d\vec{l}$ is $q\vec{E} \cdot d\vec{l}$. Dividing by q and adding up along a path:

$$V_b - V_a = - \int_a^b \vec{E} \cdot d\vec{l}.$$

The electric field is conservative, so the result is the same for every path from a to b .

Uniform field (for example, between parallel plates). Moving a distance d along the field:

$$\Delta V = -Ed, \quad \text{so} \quad E = \frac{\Delta V}{d}.$$

This is why the field can equally be given in V/m: $1 \text{ V/m} = 1 \text{ N/C}$.

Infinite line of charge. Integrating $E = 2k\lambda/r$ from r_a to r_b :

$$V_b - V_a = -2k\lambda \ln \frac{r_b}{r_a}.$$

For an infinite line we cannot put $V = 0$ at infinity, because the logarithm diverges. Only differences are meaningful.

WORKED EXAMPLE 15.4

An electron between parallel plates

Two parallel plates 5.0 mm apart are connected to a 12 V battery. An electron is released from rest at the negative plate. Find the field between the plates, and the electron's energy and speed when it reaches the positive plate.

Solution

$$E = \frac{\Delta V}{d} = \frac{12}{0.0050} = 2.4 \times 10^3 \text{ V/m}.$$

The electron moves from low to high potential and gains $K = e(12 \text{ V}) = 12 \text{ eV} = 1.9 \times 10^{-18} \text{ J}$.

$$v = \sqrt{\frac{2K}{m_e}} = \sqrt{\frac{2(1.9 \times 10^{-18})}{9.11 \times 10^{-31}}} = 2.1 \times 10^6 \text{ m/s}.$$

The plate separation does not affect the final energy. Only the potential difference matters.

§15.5

Field from the potential and equipotential surfaces

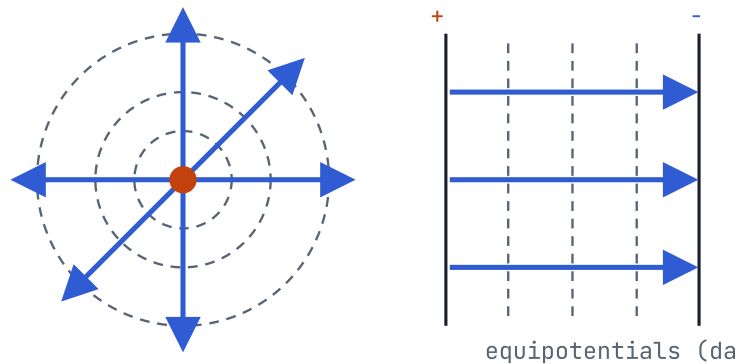
Reversing $dV = -\vec{E} \cdot d\vec{l}$ gives the field from the potential:

$$E_x = -\frac{\partial V}{\partial x}, \quad E_y = -\frac{\partial V}{\partial y}, \quad E_z = -\frac{\partial V}{\partial z}; \quad \vec{E} = -\nabla V.$$

The field points in the direction in which V decreases fastest, and its magnitude is the rate of that decrease. Like a ball on a hillside, a positive charge feels a force “downhill” on the potential map.

An **equipotential surface** is a surface on which V is constant. Moving a charge along one takes no work, so:

- Field lines cross equipotentials at right angles.
- Equipotentials are crowded together where the field is strong, just like contour lines on a steep slope.
- The surface of a conductor in equilibrium is an equipotential, and so is its whole volume.



Equipotentials (dashed) are always perpendicular to field lines. Around a point charge they are concentric spheres, crowded together where the field is strong. In a uniform field they are evenly spaced parallel planes.

WORKED EXAMPLE 15.5

Field from a potential function

In a certain region, $V(x, y) = 3x^2y - y^3$ (in volts, with x and y in metres). Find the electric field at the point $(1, 2)$.

Solution

$$E_x = -\frac{\partial V}{\partial x} = -6xy, \quad E_y = -\frac{\partial V}{\partial y} = -(3x^2 - 3y^2).$$

At $(1, 2)$: $E_x = -12$ V/m and $E_y = -(3 - 12) = +9$ V/m.

$$\vec{E} = -12\hat{i} + 9\hat{j} \text{ V/m}, \quad |\vec{E}| = 15 \text{ V/m}.$$

WORKED EXAMPLE 15.6

Checking the ring

Use the potential of a ring, $V = kQ/\sqrt{x^2 + R^2}$, to find the field on its axis.

Solution

$$E_x = -\frac{dV}{dx} = -kQ \cdot \left(-\frac{1}{2}\right)(x^2 + R^2)^{-3/2}(2x) = \frac{kQx}{(x^2 + R^2)^{3/2}}.$$

This matches the result of Chapter 13, obtained here without any vector components. Calculating V first and then differentiating is often the easiest way to find a field.

§15.6

Conductors and the potential

A conductor in equilibrium is an **equipotential**. When two conductors are connected by a wire, charge flows between them until they reach the **same potential**.

WORKED EXAMPLE 15.7

Why charge crowds onto sharp points

Two conducting spheres, of radii 1.0 cm and 10 cm, are far apart but connected by a thin wire. A total charge of 22 nC is shared between them. Find the charge on each sphere, and compare the electric fields at their surfaces.

Solution

Equal potentials (the spheres are far apart, so each has $V = kq/R$):

$$\frac{kq_1}{R_1} = \frac{kq_2}{R_2} \Rightarrow \frac{q_1}{q_2} = \frac{R_1}{R_2} = \frac{1}{10}.$$

With $q_1 + q_2 = 22$ nC: $q_1 = 2.0$ nC and $q_2 = 20$ nC.

Surface fields. $E = kq/R^2 = V/R$. Both spheres have the same V , so

$$\frac{E_1}{E_2} = \frac{R_2}{R_1} = 10.$$

The **small sphere has ten times the field**, even though it holds only one-tenth of the charge.

Evaluate. This explains why the field, and the surface charge density, are greatest at sharply curved parts of a conductor. A sharp point behaves like a tiny sphere. It is the

reason for corona discharge, lightning rods, and the smooth, rounded fittings used on high-voltage equipment.

IN PRACTICE

Potentials in the body. Every living cell maintains a potential difference across its membrane. A resting nerve or muscle cell has its interior at about -70 mV relative to the outside. The membrane is only about 7–8 nm thick, so the field across it is roughly 10^7 V/m (Problem P15.12), stronger than the field that makes air spark. A nerve impulse (action potential) is a wave in which this potential briefly flips to about $+30$ mV as sodium ions rush in, and then recovers. The ECG, EEG (brain) and EMG (muscle) all measure the tiny potential differences, from microvolts to millivolts, that these electrical events produce at the skin. Defibrillators use a large potential difference to reset the heart's electrical activity all at once.

§15.7

Potential energy and accelerators

A particle of charge q that falls through a potential difference ΔV gains kinetic energy $|q\Delta V|$. That is the whole principle of electrostatic accelerators.

- **X-ray tubes** accelerate electrons through 20–150 kV. When the electrons hit a metal target, they produce X-rays with energies up to $e\Delta V$ (Problem P15.18).
- **Electron microscopes** use 100–300 kV.
- **Medical linear accelerators** for radiotherapy reach 6–20 MeV, using oscillating rather than static fields.

WORKED EXAMPLE 15.8

Rutherford's closest approach

In Rutherford's famous experiment, alpha particles with a kinetic energy of 5.0 MeV were fired at a thin gold foil. How close can an alpha particle get to a gold nucleus ($Z = 79$) in a head-on collision? Treat the nucleus as fixed.

Solution

At the closest approach, all the kinetic energy has become electric potential energy:

$$K = \frac{k(2e)(79e)}{d} \Rightarrow d = \frac{k(158)e^2}{K}.$$

$$d = \frac{(8.99 \times 10^9)(158)(1.602 \times 10^{-19})^2}{5.0 \times 1.602 \times 10^{-13}} = \frac{3.65 \times 10^{-26}}{8.0 \times 10^{-13}} = 4.6 \times 10^{-14} \text{ m} = 46 \text{ fm}.$$

Evaluate. This is about 1/1000 of the size of an atom. Rutherford concluded (1911) that the atom's positive charge is concentrated in a tiny nucleus, less than about 10^{-14} m across. We now know the gold nucleus has a radius of about 7 fm, so these alpha particles never actually touched it.

CHAPTER SUMMARY

- Potential energy of two point charges: $U = kq_1q_2/r$. For a system, sum over all pairs.
- Potential $V = U/q_0$, in volts. $\Delta U = q\Delta V$. Positive charges move towards lower V .
 $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$.
- Point charge: $V = kq/r$. Potentials add as scalars: $V = k \int dq/r$. Ring on its axis:
 $kQ/\sqrt{x^2 + R^2}$.
- $V_b - V_a = - \int_a^b \vec{E} \cdot d\vec{l}$. In a uniform field, $E = \Delta V/d$.
- $\vec{E} = -\nabla V$. Field lines are perpendicular to equipotentials and point towards lower V .
- Conductors are equipotentials. Connected conductors share a common potential, so smaller radius means larger surface field.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P15.1

(a) If the electric field is zero at a point, must the potential be zero there? (b) If the potential is zero at a point, must the field be zero there? Give an example for each.

P15.2

Explain why a bird can safely perch on a high-voltage line, but would be electrocuted if it touched two lines at once.

P15.3

Explain why electric field lines must always cross equipotential surfaces at right angles.

P15.4

An electron moves from a point at low potential to a point at high potential. Does its electric potential energy increase or decrease? Does its kinetic energy increase or decrease, if no other forces act?

LEVEL B – STANDARD PROBLEMS**P15.5**

Find the work needed to assemble four charges of $+1.0 \mu\text{C}$ at the corners of a square of side 10 cm, starting from far apart.

P15.6

For the arrangement of P15.5, find the potential and the electric field at the centre of the square.

P15.7

An alpha particle (charge $+2e$, mass $6.64 \times 10^{-27} \text{ kg}$) is accelerated from rest through 2.0 MV. Find its kinetic energy in MeV and its speed.

P15.8

An alpha particle of kinetic energy 7.7 MeV is fired head-on at a gold nucleus ($Z = 79$). Find its distance of closest approach. How does this compare with the nuclear radius of about 7 fm?

P15.9

Two parallel plates 2.0 cm apart have a potential difference of 500 V. A proton is released from rest at the positive plate. Find the field between the plates and the proton's speed when it reaches the negative plate.

P15.10

A ring of radius 10 cm carries 10 nC spread uniformly. Find the potential at its centre and on its axis at 10 cm from the centre. How much work is needed to bring a 1.0 nC charge from far away to the centre?

P15.11

The potential along the x -axis is $V(x) = 5x^2 - 3x$ (in volts, with x in m). Find E_x as a function of x , the field at $x = 2.0$ m, and the point where a charged particle would be in equilibrium.

P15.12

Nerve membrane. A resting nerve cell has its interior at -70 mV relative to the outside, across a membrane 8.0 nm thick. (a) Find the field in the membrane, treating it as uniform. (b) How much energy does a sodium ion ($+e$) gain in crossing from outside to inside? Express it in eV and J, and compare it with the thermal energy kT at body temperature (310 K).

LEVEL C – CHALLENGE PROBLEMS**P15.13**

Find the potential on the axis of a uniformly charged disc (charge density σ , radius R) by summing ring potentials. Show that $V = \frac{\sigma}{2\epsilon_0} (\sqrt{x^2 + R^2} - x)$, then differentiate to recover the field found in Chapter 13.

P15.14

A solid insulating sphere of radius R carries charge Q spread uniformly. (a) Using $V = -\int \vec{E} \cdot d\vec{l}$ from infinity, show that inside the sphere $V(r) = \frac{kQ(3R^2 - r^2)}{2R^3}$. (b) Show that the potential at the centre is 1.5 times the potential at the surface.

P15.15

The energy of a charged sphere. (a) By building a uniformly charged sphere up one thin shell at a time, show that the energy needed to assemble it is $U = \frac{3kQ^2}{5R}$. (b) Estimate the electrostatic energy of a uranium nucleus ($Z = 92$, $R = 7.4$ fm) in MeV. (c) In fission, the nucleus splits into two smaller nuclei, each with about half the charge and a radius $2^{-1/3}$ times smaller. Estimate the change in electrostatic energy, and compare it with the roughly 200 MeV released per fission.

P15.16

A dipole $\vec{p}$ points along the z -axis. (a) Show that at distances $r \gg d$, its potential is $V = \frac{kp \cos \theta}{r^2}$, where θ is measured from the z -axis. (b) Using $E_r = -\partial V / \partial r$ and $E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta}$, find the field components, and check them against the axial and equatorial results of Chapter 13.

P15.17

A coaxial cable has an inner conductor of radius $a = 0.50$ mm and an outer conductor of radius $b = 3.0$ mm. (a) Show that the potential difference between them is $\Delta V = 2k\lambda \ln(b/a)$. (b) What charge per metre gives a potential difference of 1.0 kV? (c) What is the largest field in the cable at this voltage?

P15.18

X-ray tube. A medical X-ray tube accelerates electrons through 100 kV. (a) Find their kinetic energy in keV and in joules. (b) Calculate their speed using $K = \frac{1}{2}mv^2$, and comment on the answer. (c) Using the relativistic result $K = (\gamma - 1)m_e c^2$, with $m_e c^2 = 511$ keV and $\gamma = 1/\sqrt{1 - v^2/c^2}$, find the actual speed. (d) The most energetic X-ray photon produced carries all of an electron's kinetic energy. Using $E = hc/\lambda$, with $hc = 1240$ eV nm, find the shortest X-ray wavelength produced.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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